

Q1

Directions for question 1: Type in your answer in the input box provided below the question.

A train travels 100 metres in 20 seconds. What is its speed in kmph?

Solution:

$$\frac{100}{20} \frac{\text{m}}{\text{s}} = 300 \frac{\text{m}}{\text{min}} = 18000 \frac{\text{m}}{\text{hr}} = 18 \frac{\text{km}}{\text{hr}}$$

Ans : (18)

Q2



Directions for questions 2 to 4: Select the correct alternative from the given choices.

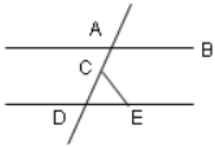
A can do a piece of work in $2a$ days, B in a days and C in $4a$ days. How long will they take to complete the work if the three work together?

☒ a) $\frac{4a}{7}$ days (Your answer - ✓)

- ☐ b) $6a$ days
- ☐ c) $2a$ days
- ☐ d) None of these

Q3

Directions for questions 2 to 4: Select the correct alternative from the given choices.



In the figure, AB and DE are parallel. If

$\angle$

$BAD = 110^\circ$ and

$\angle$

$DEC = 30^\circ$, find

$\angle$

DCE.

☒ a) 80° (Your answer - ✓)

☐ b) 70°

☐ c) 60°

☐ d) 50°

Q4

Directions for questions 2 to 4: Select the correct alternative from the given choices.

Find the ratio of the total surface area of a sphere to that of a hemisphere whose respective radii are in the ratio of 2 : 3.

☐ a) 16 : 27

☐ b) 1 : 1

☒ c) 2 : 1 (Your answer - ✖)

☐ d) 8 : 9

Solution:

The ratio of the radii of the sphere and hemisphere is 2 : 3, say the radii are 2 and 3 respectively.

The total surface area of the sphere is $4\pi(2^2) = 16\pi$.

The total surface area of the hemisphere is $3\pi(3^2) = 27\pi$. The required ratio is 16 : 27.

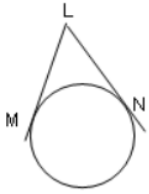
Choice (A)

Directions for question 5: Type in your answer in the input box provided below the question.

A can do a work in 15 days. B can do the same in 20 days. They start the work together and A leaves after three days. In how many days is the work completed?

Your Answer: 16 ✓

Directions for questions 6 and 7: Select the correct alternative from the given choices.



In the figure, LM and LN are tangents to the circle. If

$\angle$

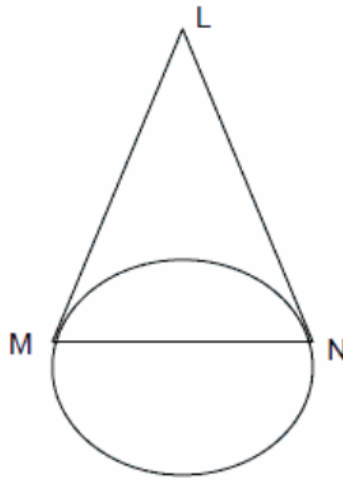
$MLN = 50^\circ$, find

$\angle$

LMN.

- ☐ a) 65°
- ☐ b) 75°
- ☐ c) 55°
- ☐ d) 35°

Solution:



$\angle MLN = 50^\circ \therefore \angle LNM = \angle LMN = 65^\circ$
($\because$ isosceles triangle)

Choice (A)

Directions for questions 6 and 7: Select the correct alternative from the given choices.

A train having a length of 400 m, travelling at a speed of 20 m/s, crosses a platform in one minute. What is the length of the platform?

- ☐ a) 2000 m
- ☒ b) 800 m (Correct Answer - ✓)
- ☐ c) 400 m
- ☐ d) Cannot be determined

Directions for question 8: Type in your answer in the input box provided below the question.

Walking at 40% more than my usual speed, I reach my office six minutes early. Find my usual time (in minutes) to reach my office.

Your Answer: 15 ✖

Correct Answer: 21 ✔

Solution:

Let my normal speed be $5u$ and the normal time be $7t$. If I walk at a speed which is 40% greater, i.e. at $7u$, I would take $5t$. As $7t - 5t = 6$ min, $t = 3$ min. The normal time, $7t = 21$ min.
Ans : (21)

Problem Setup

- Usual speed = S
- Increased speed = $1.4S$ (since 40% more)
- Usual time = T minutes
- Early arrival = 6 minutes

Distance Relation

Distance is constant, so:

$$S \cdot T = 1.4S \cdot (T - 6)$$

Cancel S :

$$T = 1.4(T - 6)$$

Simplify

$$T = 1.4T - 8.4$$

$$1.4T - T = 8.4$$

$$0.4T = 8.4$$

$$T = 8.4 \cdot 0.4 = 21$$

Q9



Directions for question 9: Select the correct alternative from the given choices.

A can do a work in 6 hours while B and C can do the same work in 8 hours and 12 hours respectively. How long will they take to complete the work together?

- ☐ a) $2\frac{1}{2}$ hours
- ☐ b) $2\frac{3}{4}$ hours
- ☐ c) 2 hours

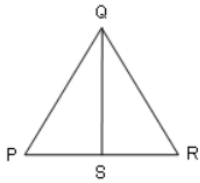
☒ d) None of these (Your answer - ✓)

Solution:

A, B, C take 6, 8, 12 hours respectively. Their rates are 4, 3, 2 units/day (if the work is taken as 24 units) respectively. Their combined rate is 9 units/day. They would take $\frac{24}{9}$, i.e. $\frac{8}{3}$, i.e. $2\frac{2}{3}$ days to complete the work. Choice (D)

Q10

Directions for question 10: Type in your answer in the input box provided in the question.



In the figure,

$\angle$

$QSR = 6$

$\angle$

$SQR,$

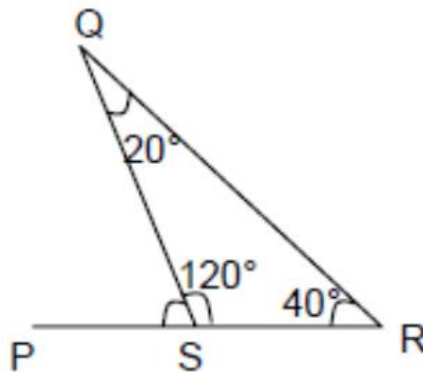
$\angle$

$QSP = 60^\circ$. Find

$\angle$

QRS (in degrees).

Solution:



$$\angle S = 60^\circ \therefore \angle QSR = 120^\circ$$

$$\angle QSR = 6 \angle SQR. \therefore \angle SQR = 20^\circ$$

$$\therefore \angle R = 40^\circ.$$

Ans : (40)

Directions for question 11: Select the correct alternative from the given choices.

Two circles which touch each other externally have the ratio of their areas as 4 : 9. What is the radius of the bigger circle, if the distance between the centres of the two circles is 75 cm?

☐ a) 30 cm

☒ b) 45 cm (Correct Answer - ✓)

☐ c) 60 cm

☐ d) 40 cm

Solution:

The ratio of the areas of the two circles is 4 : 9.

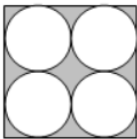
∴ The ratio of their radii is 2:3. The sum of the radii is 75 cm.

∴ The radii are 30 cm and 45 cm.

Choice (B)

Directions for question 12: Type in your answer in the input box provided below the question.

In the given figure, find the area of the shaded region (in sq. cm), if the radius of each of the circles drawn inside the square is 7 cm.



Your Answer: 168 ✓

Correct Answer: 168 ✓

Solution:

The side of the square is 28 cm.

∴ Its area is 784 cm^2 . The total area of the 4 circles is $4\pi(7^2) \text{ cm}^2 = 4 \left(\frac{22}{7}\right) (7^2) \text{ cm}^2$.
 $= 616 \text{ cm}^2$.

The area of the shaded region is $(784 - 616) \text{ cm}^2$
 $= 168 \text{ cm}^2$.

Ans : (168)

Directions for question 13: Select the correct alternative from the given choices.

Five men undertake a job to dig a well 25 m deep in 10 days. 10 boys can fill up such a well 5 m deep in 20 days. If both the groups start the work together, then after how many days is the well completed?

☐ a) $9\frac{1}{10}$

☐ b) $9\frac{1}{11}$

☒ c) $11\frac{1}{9}$ (Correct Answer - ✓)

☐ d) $10\frac{1}{9}$

Step 1: Work of the men

- 5 men dig a 25 m well in 10 days. So, their rate = $\frac{25}{10} = 2.5 \text{ m/day}$.

Step 2: Work of the boys

- 10 boys fill a 5 m well in 20 days. So, their rate = $\frac{5}{20} = 0.25 \text{ m/day}$.

Step 3: Net work rate

Since the boys are filling while the men are digging:

$$\text{Net rate} = 2.5 - 0.25 = 2.25 \text{ m/day}$$

Step 4: Total time

Well depth = 25 m. So,

$$Time = 252.25 = 1009 \approx 11.11 \text{ days}$$

Correct Answer

$$1009 \text{ days} \approx 11.1 \text{ days}$$

That matches the highlighted choice **1009 (≈ 11.1 days)**.

Q14



Directions for question 14: Type in your answer in the input box provided below the question.

A bus travels from A to B at a speed of 65 kmph in one hour. If the speed is reduced by 15 kmph, how many more minutes will it take to cover the same distance?

Problem Setup

- Speed from A to B = **65 km/h**
- Time taken = **1 hour** $\rightarrow$ Distance = *65 km*
- Reduced speed = $65 - 15 = 50 \text{ km/h}$

Calculation

New time:

$$Time = \frac{Distance}{Speed} = \frac{65}{50} = 1.3 \text{ hours}$$

Convert to minutes:

$$1.3 \times 60 = 78 \text{ minutes}$$

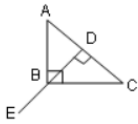
Difference:

$$78 - 60 = 18 \text{ minutes}$$

☒ **Correct Answer**

It will take **18 minutes more**.

Directions for question 15: Select the correct alternative from the given choices.



In the figure,

$\angle$

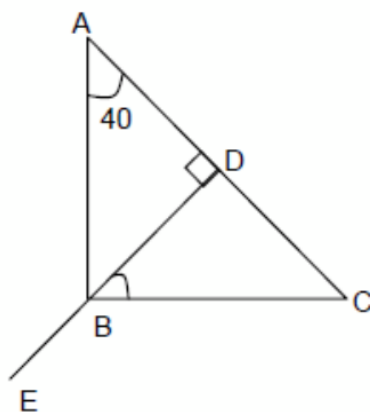
$BAC = 40^\circ$, find

$\angle$

CBE .

- ☐ a) 110°
- ☐ b) 120°
- ☐ c) 130°
- ☐ d) 140°

Solution:



$$\begin{aligned}\angle A = 40^\circ &\Rightarrow \angle ABD = 180^\circ - (40^\circ + 90^\circ) = 50^\circ \\ \angle DBC = 40^\circ &\Rightarrow \angle CBE = 140^\circ\end{aligned}$$

Choice (D)